Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr.S60(N&S), Elite, IAS & LIIT | JEE-MAIN-Shift-2 | Date: 19-03-2026

Time: 03.00Pm to 06.00Pm | GTM-24 | Max. Marks: 300

KEY SHEET

MATHEMATICS

1	**2**	2	**1**	3	**4**	4	**2**	5	**1**
6	**1**	7	**3**	8	**1**	9	**2**	10	**2**
11	**4**	12	**1**	13	**4**	14	**4**	15	**3**
16	**1**	17	**3**	18	**2**	19	**3**	20	**2**
21	**5**	22	**15**	23	**182**	24	**362**	25	**21**

PHYSICS

26	**1**	27	**4**	28	**2**	29	**1**	30	**1**
31	**2**	32	**1**	33	**4**	34	**2**	35	**3**
36	**1**	37	**2**	38	**2**	39	**4**	40	**4**
41	**1**	42	**3**	43	**2**	44	**4**	45	**3**
46	**5**	47	**9**	48	**5**	49	**23**	50	**34**

CHEMISTRY

51	**2**	52	**4**	53	**2**	54	**3**	55	**1**
56	**4**	57	**4**	58	**1**	59	**1**	60	**1**
61	**3**	62	**4**	63	**3**	64	**2**	65	**4**
66	**1**	67	**1**	68	**1**	69	**2**	70	**2**
71	**287**	72	**5**	73	**50**	74	**416**	75	**13**

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SOLUTIONS
MATHEMATICS

1. $f(3)<f(3^-)\Rightarrow -15<12-27+\ell n(a^2-3a+3)\Rightarrow a^2-3a+2>0\Rightarrow (a-1)(a-2)>0$

2. $T_{r+1}={}^{15}C_r\left(2^{1/5}\right)^{15-\ell}\qquad r=0,15$

3. $x^2-1\geq -[x]$ from the graphs

4. Equation of normal $y+xt=2at+at^3 T(6,0)\Rightarrow t^2=4\qquad P(t)=(4,4)A,$

   No common tangent $r<AC\qquad\therefore r<\sqrt{20}$

5. A.P $d_1=\frac{3}{4},a=-129\frac{1}{4}$

   Required term $=129\frac{1}{4}+19.\frac{3}{4}\qquad=-115$

6. Foot $=(1,3,5)\Rightarrow$ image $=(1,6,3)$

   $\alpha=30^o\qquad\bar{r}=\left(\hat{i}+6\hat{j}+3\hat{k}\right)+t\left(\frac{\hat{i}}{2}-\frac{1}{2}\hat{j}\frac{-1}{\sqrt{2}}\hat{k}\right)$

   By verification option

7. AB = AC $\therefore D$ is midpoint of BC

   $D=\left(\frac{1}{2},\frac{3}{2},3\right)\therefore \ell=AD=\sqrt{\frac{42}{2}}\qquad 2\ell^2=45$

8. $\cos\theta=\frac{1}{3}\qquad\therefore n=13$

   $S=\frac{1}{2}+\frac{2}{2^2}+.....+\frac{13}{2^{13}}$

   $\sum_{k=1}^{13}\frac{k}{2^k}\qquad=\frac{\frac{1}{2}S=\frac{1}{2^2}+2\frac{1}{2^3}+.....+\frac{13}{2^{14}}}{\frac{S}{2}=\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....+}$

   $S=\frac{2^{14}-15}{2^{13}}$

9. $\underset{x\to 0}{Lt}\frac{\int_0^x f(t)dt}{x}=\underset{x\to 0}{Lt}f(x)=f(0)=\frac{1}{2}$

10. For unique $\Delta\neq 0.\ \lambda\neq\frac{1}{2}$

    For no sol $\Delta=0,\Delta_x=0\Rightarrow m=15,1$

    For ∞ No, sols $\lambda=\frac{1}{2},\mu=1(or)15$

11. $M=\ell+\frac{\frac{N}{2}-C}{f}\times 4\qquad=8+\frac{18-12}{10}\times 4=10.4$

12. $R_2\to R_2-R_1,R_3\to R_3-R_1$

    $f(x)=45\Rightarrow f^1(x)=0$

13. $I_2=\int_0^1\frac{x^2}{e^{x^3}\left(2-x^3\right)}dx\ x^3=t\qquad=\frac{1}{3}\int_0^1\frac{dt}{e^t(2-t)},1-t=z\ =\frac{1}{3e}I_1$

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14. $\text{Re}(z) = |z-2|$

$$\left.\begin{aligned} 4x_1 &= y_1^2 + 4 \\ 4x_2 &= y_2^2 + 4 \end{aligned}\right\} \to (I)$$

$$\frac{y_1 - y_2}{x_1 - x_2} = \sqrt{3} \to (II) \qquad \therefore y_1 + y_2 = \frac{4}{\sqrt{3}}$$

15. center $\left(\sqrt{10}, 3\right)$ radius $= \sqrt{10}$

$$L.C = 2\sqrt{r^2 - d^2} = 2\sqrt{10}$$

16. $e = \sqrt{\dfrac{a^2 - b^2}{a^2}}$

17. $S_1 : (1,2) \in S$ but $(2,1) \in S$

$S_2 : x - x = 0, \in I, x - y \in I \qquad \Rightarrow y - x \in II$

$x - y, y - z \in I \Rightarrow x - z \in I$

18. $I = 2x - 4\,\text{Tan}^{-1}\dfrac{x}{2} + c, c = 0$

$f(x) = 4\,\text{Tan}^{-1}\left(\dfrac{x}{2}\right) \qquad \therefore \min = -\pi$

19. Area of $\Delta oAB = 1$

Required area $< 1 \quad \therefore [k] = 0$

20. Linear D.E $\displaystyle\int \frac{1}{2\cos^4 x - \cos 2x}\,dx = \int \frac{1}{c^4 + s^4}\,dx = -\text{Tan}^{-1}\left(\sqrt{2}cot 2x\right)$

$y.(I.F) = \dfrac{x^2}{2} + c, c = 0$

21. $D < 0, 64 - 4(n-2)(n-4) < 0 \Rightarrow n > 4$

22. $$f(x) = \begin{cases} \dfrac{1}{x} & x \geq 2 \\ -\dfrac{1}{x} & x \leq -2 \\ ax^2 + 2b & -2 < x < 2 \end{cases}$$

Is continuous at x = 2 $\quad \dfrac{1}{2} = \dfrac{a}{4} + 2b$

Differentiable at $x = 2 \quad \Rightarrow \dfrac{-1}{4} = 4a \qquad \therefore a = \dfrac{-1}{16},\ b = \dfrac{3}{8}$

23. $\tan 30^o = \dfrac{b^2}{a^2 e} \Rightarrow b^4 - 3b^2 - 27 = 0 \qquad \Rightarrow b^2 = \dfrac{3}{2}\left(1 + \sqrt{13}\right)$

24. $D_4 + D_5 + D_5 + D_6 = 362$

25. A : cons shows no heads

$B_1 : 1(or)2 \qquad B_2 : 3 \qquad B_3 : 5(or)6$

$$P(B_1 / A) = \frac{P(B_1)\ \ p(A/B_1)}{\sum_{i=1}^{3} p(B_i) P(A/B_i)} = \frac{8}{13} = \frac{m}{n}$$

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PHYSICS

26. Nuclear Reactor

27. Both assertion & reason are true and the reason is correct explanation of assertion.

28. $\chi = C.\frac{1}{T} \Rightarrow C = \chi.T = \frac{0.4}{7\times10^{-3}} = 57K$

29. $P_{ex} = \frac{4T}{r}$

30. $\bar{E}\times\bar{B}$ is along positive z-direction, $\bar{B}$ is along positive $\bar{J}$ direction

31. $\beta_{max} = \frac{\lambda D}{d_0 - a_0}$ $\beta_{min} = \frac{\lambda D}{d_0 + a_0}$

32. $a_x = \frac{dv_x}{dt} = \alpha\frac{dy}{dt} = \alpha.2\alpha x = 2\alpha^2 x$ $a_y = \frac{dv_x}{dt} = 2\alpha\frac{dy}{dt} = 2\alpha.y\alpha = 2\alpha^2 y$

Force, $\vec{F} = m\vec{a} = m\left(a_x\hat{x} + a_y\hat{y}\right)$

$\vec{F} = m\left(2\alpha^2 xx + 2\alpha^2 y\hat{y}\right)$ $\vec{F} = 2m\alpha^2\left(x\hat{x} + y\hat{y}\right)$

33. Fundamental frequency f_o =50Hz

$f_o = \frac{V}{4l}$ $50 = \frac{300}{4l} \Rightarrow l = 1.5m$

34. $U = \frac{1}{2}CV^2 = \frac{1}{2}\left(\frac{\varepsilon_0 A}{d}\right)(Ed)^2 = \frac{\varepsilon_0 AE^2 d}{2}$ $u = \frac{U}{V} = \frac{\varepsilon_0 AE^2 d}{Ad} = \frac{1}{2}\varepsilon_0 E^2$

35. $V_{rms} = \sqrt{V_R^2 + (V_L - V_C)^2} = \sqrt{(40)^2 + (40-10)^2} = 50V$

$i_{rms} = \frac{iv}{\sqrt{2}} = \frac{10\sqrt{2}}{\sqrt{2}} = 10A$ $z = \frac{V_{rms}}{i_{rms}} = \frac{50}{10} = 5\Omega$

36. $W = \frac{1}{2}mv^2$ $Vq = \frac{1}{2}mv^2$ $\lambda = \frac{h}{p} = \frac{h}{mv} = \frac{h}{\sqrt{2mVq}}$ $\therefore \lambda p = \frac{h}{\sqrt{2mVe}}$

$\lambda_\alpha = \frac{h}{\sqrt{2(4m)(2e)V}} = \frac{\lambda p}{2\sqrt{2}}$ $\therefore \frac{\lambda p}{\lambda_\alpha} = 2\sqrt{2}$

37. $mA\omega^2 \geq mg$ $\omega = \sqrt{\frac{g}{A}}$ $T = 2\pi\sqrt{\frac{A}{g}}$

38. $h = \frac{w^2x^2}{2g} = \frac{(2\pi\times2)^2(0.05)^2}{2g} = 2cm$

39. $\frac{1}{f} = (1.5-1)\frac{2}{R}$

For water lens $\frac{1}{f_w} = \left(\frac{4}{5}-1\right)\left(\frac{-2}{R}\right) = \frac{-2}{3f}$

$\therefore \frac{1}{p} = \frac{2}{f} + \frac{1}{f_w} = \frac{2}{f} - \frac{2}{3f} = \frac{4}{3f}$ $F = \frac{3f}{4}$

40. $R = \frac{\rho l}{A}$; $R_1 = \frac{\rho l_1}{A_1} = \frac{\rho\left(\frac{1}{2}\right)}{\pi(2r)^2}$ $R_2 = \frac{\rho l_2}{A_2} = \frac{\rho\left(\frac{1}{2}\right)}{\pi r^2}$ $\therefore \frac{R_1}{A_2} = \frac{1}{4}$ $\frac{J_1}{J_2} = \frac{I}{A_1}\times\frac{A_2}{I} = \frac{1}{4}$

Voltage drop $V = iR$; $\frac{V_1}{V_2} = \frac{iR_1}{iR_2} = \frac{R_1}{R_2} = \frac{1}{4}$

Power drop $P = i^2R$ $\frac{p_1}{p_2} = \frac{i^2R_1}{i^2R_2} = \frac{1}{4}$

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41. Process is isobaric, (p=count) expansion, hence temperature of the gas will increase
Process-2 is an isothermal, hence, here $\Delta u_2 = 0$
Process-3 is an adiabatic expansion. Hence temperature of the gas will increase
$\therefore \Delta u_1 > \Delta u_2 > \Delta u_3$

42. Height of centre of mass of given chain
$h_1 = \frac{l}{2};\ h_2 = \frac{l}{2};\ h_3 = \frac{2r}{\pi} = \frac{2L}{\pi^2}$ $h_4 = \frac{2l}{\theta}\sin^2\left(\frac{\theta}{2}\right) = \frac{2l}{\pi};\ h_5 = r - \frac{2r}{\pi} = \frac{l}{\pi} - \frac{2l}{\pi^2}$ and $U = mgh$

43. To compensate for the change in length caused by changes in temperature of atmosphere during expt.

44. $i = \frac{E}{\text{Re } ff} = \frac{6}{50+130+120} = \frac{6}{300} = 20mA$

45. The potential due to this pair at any point on y-axis is zero. The sum of potential due to all such possible pairs is zero at all points on y-axis. Hence potential at $p\left(0, \frac{R}{2}\right)$ is zero.

46. $M = \int_0^R \rho 4\pi r^2 dr = \int_0^R Kr 4\pi r^2 dr = 4\Pi K \int_0^R r^3 dr = \frac{4\pi KR^4}{4} = \pi KR^4$ $V_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{G.\pi KR^4}{2R}} = \sqrt{\frac{\pi KGR^3}{2}}$

47. Right side acceleration of the blocks $a = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)g = \left(\frac{2-1}{2+1}\right)10$ $a = \frac{10}{3} m/s^2$

Tension in pulley $T_p = 2\left(\frac{2m_1m_2}{m_1+m_2}\right)g = \frac{4\times1\times2}{3}\times10 = \frac{80}{3}N$

At equilibrium $\sum\tau = 0$

$l\times30 = (1-l)\frac{80}{3}$ $9l = 8-8l$ $17l = 8$ $l = \frac{8}{17}$

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48. If velocity of rod v then induced emf $e = B/v$

Change on capacitor q=Ce=CB/V $i = \frac{dq}{dt} = CBl\left(\frac{dv}{dt}\right)$

Net force on rod $ma = mg - Bil$

$m\frac{dv}{dt} = mg - B\left(CBl\frac{dv}{dt}\right)l$ $m\frac{dv}{dt} + B^2l^2C\frac{dv}{dt} = mg$

$\frac{dV}{dt}\left(m + B^2l^2C\right) = mg$ $\frac{dv}{dt} = \frac{mg}{m + B^2l^2C}$

$= \frac{1\times10}{1+1^2(0.5)^2\times4}$ $= \frac{10}{2}$ $\frac{dv}{dt} = 5m/s^2$

49. $P = \frac{2NMv\cos\theta}{A}$ $= \frac{2\times10^{23}\times3.23\times10^{-27}\times10^3}{2\times10^{-4}}\times\frac{1}{\sqrt{2}}$

$= \frac{3.23\times10^3}{\sqrt{2}}$ $= 2.30\times10^3$ $= 23\times10^2\ pa$

50. $d = P.S.R + L.S.R\times L.C$ $= 1.5mm + 7\times\frac{0.5}{50}mm$ $= 1.57\text{mm}$

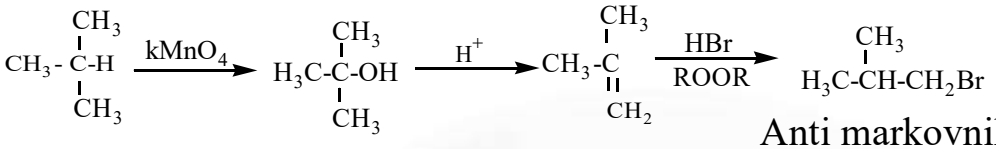
$S = 2\pi rL = 1.57\times10^{-3}\times2\times3.14\times6.8\times10^{-2} = 34\times10^{-5}m^2$

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CHEMISTRY

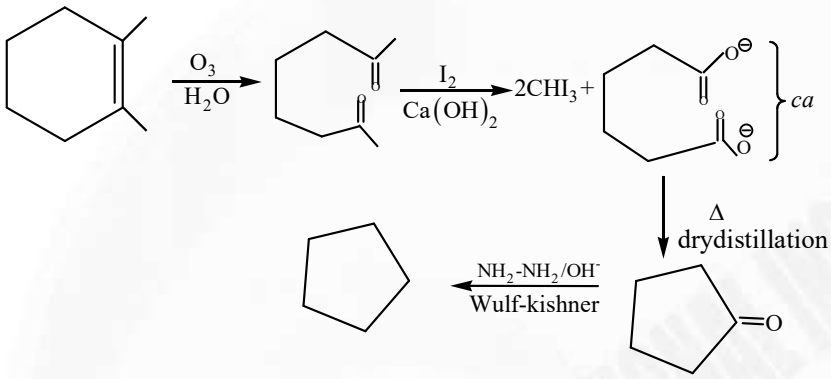
51. There is no 'c' for formation of NaCN

52.



Anti markovnikoff

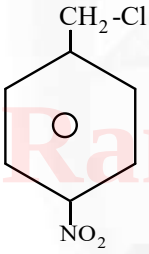
53.



54. In Gabriel phthalimide Synthes is only aliphatic primary amines

55. Acidic strength increased by -I groups

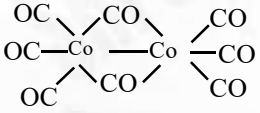
56.



It is most reactive to wards SN^2 because it has most stable transition state

57. All amino acids except Glycine are optically active

58.



59. P) $[FeF_6]^{3-}$

Fe^{3+}-$3d^5$-t_{2g}^3 e_g^2

$\mu=\sqrt{n(n+2)}BM$

$\mu=\sqrt{5(5+2)}=\sqrt{35}BM$

Q) $\left[V(H_2O)_6\right]^{2+}$

$V^{+2}-3d^3-t_{2g}^3\ e_g^0$

$\mu=\sqrt{3(3+2)}=\sqrt{15}BM$

R) $\left[Fe(H_2O)_6\right]^{2+}$

Fe^{+2}-$3d^6$-t_{2g}^4 e_g^2

$\mu=\sqrt{4(4+2)}=\sqrt{24}BM$

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60. $\sigma 1s^2\sigma^* 1s^2\sigma 2s^2\sigma^* 2s^2\sigma 2p_z^2\pi 2p_x^2 = \pi 2p_y^2\pi^* 2p_x^1 = \pi^* 2p_y^1\sigma^* 2p_z$

61. 1) Mn_2O_7: Multiple bonds

2) $\overset{+4}{VO_2^+} < \overset{+6}{Cr_2O_7^{-2}} < \overset{+7}{MnO_4^-}$

3) When I^- is oxidized by MnO_4^- in alkaline medium I^- converts in to IO_3^-

4) $AgNO_3 \xrightarrow{\Delta} Ag\downarrow + NO_2\uparrow + O_2\uparrow$

62. $N(CH_3)_3$ and $P(CH_3)_3$ both are lewis base and act as ligand, However, $P(CH_3)_3$ has a $\pi-$ acceptor character

63.

A) $R\text{-}\overset{\overset{O}{\|}}{C}\text{-}H \xrightarrow[conc.HCl]{Zn-Hg} R-CH_3$

Clemmensen reduction

B) $R\text{-}\overset{\overset{O}{\|}}{C}\text{-}Cl \xrightarrow{H_2/Pd-BaSO_4} R-CHO$

Rosenmund reduction

C) $R\text{-}\overset{\overset{O}{\|}}{C}\text{-}CH_3 \xrightarrow{I_2+NaOH} RCOONa + CHI_3$

Iodoform reaction

D) $2HCHO \xrightarrow{Conc.KOH} CH_3OH + HCOOK$

Cannizzaro reaction

64. $K = \frac{2.303}{20}\log\frac{100}{90}; K = \frac{2.303}{t_{19\%}}\log\frac{100}{81}$

$\frac{2.303}{20}\log\frac{10}{9} = \frac{2.303}{t_{19\%}}\log\left(\frac{10}{9}\right)^2$

$\frac{1}{20}\log\frac{10}{9} = \frac{2}{t_{19\%}}\log\frac{10}{9}$

$\frac{1}{20} = \frac{2}{t_{19}\%}$

$t_{19\%} = 40\,min$

65. No. of spectral lines given by $H = \frac{5\times 4}{2} = 10$

No. of spectral lines given by $He^+ = \frac{4\times 3}{2} = 6$

No. of spectral lines given by $Li^{+2} = \frac{3\times 2}{2} = 3$

However $\lambda_{2\to 1}(H) = \lambda_{4\to 2}(He^+)$

$\therefore$ Total no. of spectral lines = 10+6+3-1=18

66. $\left[Ca^{+2}\right]_{final} = \frac{10^{-2}\times V}{2V} = 5\times 10^{-3}$

$\left[F^-\right]_{final} = \frac{10^{-3}\times V}{2V} = 5\times 10^{-4}$

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Ionic product $=\left[Ca^{+2}\right]\left[F^{-}\right]^2=\left(5\times10^{-3}\right)\left(5\times10^{-4}\right)^2=125\times10^{-11}=1.25\times10^{-9}$

Ionic product > k_{sp} $\left(1.7\times10^{-10}\right)$

Ppt will from

67. $Fe^{+3}+3e^{-}\rightarrow Fe \quad \Delta G_1^0=-3F(-0.04)$

$Fe\rightarrow Fe^{+2}+2e^{-} \quad \Delta G_2^0=-2F(+0.44)$

$Fe^{+3}1e^{-}\rightarrow Fe^{+2} \quad \Delta G_3^0=-1F\ E_3^o$

$\Delta G_3^0=\Delta G_1^0+\Delta G_2^0$

$-1F\ E_3^o=-3F(-0.04)-2F(0.44)$

E_3^o=0.76V

68. Cl_2, Br_2 and I_2 can undergo disproprotination Reaction

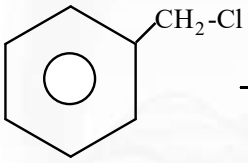
69. $x \rightleftharpoons y \quad k_1=1.0$, $y \rightleftharpoons Z \quad k_2=2.0$, $Z \rightleftharpoons w \quad k_3=4.0$

$x \rightleftharpoons w \quad k=k_1\times k_2\times k_3$

$k=1\times2\times4 \qquad k=8$

70. H_2S is a weak acid, its dissociation controlled by HCl and NH_4OH

71. P-toluene sulphonyl chloride clear solution means 3°-amine



$\xrightarrow{NH_3}$ $Ph-CH_2-N(-CH_2-Ph)-CH_2-Ph$

(x)

Molar mass=287g mol^{-1}

72. $\Delta T_b=i\,k_b\ m$

$[100.52-100]=i\times0.52\times\left(\frac{10}{200}\times\frac{1000}{100}\right)$

$0.52=i\times0.52\left(\frac{1}{2}\right) \qquad i=2$

For dissociation

$i=1+(n+1)\alpha \quad 2=1+(3-1)\alpha \quad \alpha=0.5=5\times10^{-1}$

73. g.Eq $_{\text{of Fe}}{}^{+2}$ = g.Eq of $KMnO_4$

$\frac{w}{392}\times1=0.1\times\frac{50}{1000} \qquad wt=\frac{0.1\times50\times392}{1000}$

$wt=1.96$ Purity $=\frac{1.96}{392}\times100=50\%$

74. $CH_{4(g)}\rightarrow C_{(g)}+4H_{(g)} \quad \Delta H=1665\ kJ/mol$

$C-H$ bond energy $=\frac{1665}{4}$=416.kJ mol^{-1}

75. Except $NH_{3(g)}$ and $H_2O_{(g)}$ all others have H-bonding

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